LO3 Quick Revision Guide

Based on the uploaded learning outcomes.

1. Pod with shared localhost

- Create pod: `podman pod create -p 8080:80`
- Run app and DB in same pod.
- Containers share the **network namespace**, so communication uses **localhost**.

2. User-defined network

- `podman network create appnet`

- Run both containers with `--network appnet`

- Verify:

  podman exec app ping postgres

- Podman provides DNS-based name resolution.

3. Two-tier application

Deploy any supported application (e.g. Nextcloud + MariaDB, Gitea + PostgreSQL, Ghost + MySQL) using containers and complete first-run setup.

4. Named volumes

Create:

podman volume create db_data

Mount:

-v db_data:/var/lib/mysql

Delete container, recreate it and verify the data still exists.

5. Podman secrets

podman secret create dbpass password.txt

Use:

--secret dbpass

Benefit: password is not visible in command history or environment variables.

6. podman-compose

Create `compose.yaml` and run:

podman-compose up -d

7. Internal database network

Publish only the application port. Database has **no published ports**, making it inaccessible from the host.

8. Health checks

Use: - `healthcheck` - `depends_on` - `condition: service_healthy`

Application starts only after the database becomes healthy.

9. env_file

Move credentials into:

.env

Reference:

env_file:

  - .env

10. Scaling

podman-compose up --scale app=3

Traffic is distributed among replicas (typically by a reverse proxy).

Key concepts to remember

- Pod = shared network namespace.
- Network = DNS between containers.
- Volume = persistent storage.
- Bind mount = host configuration.
- Secret = secure credentials.
- Compose = multi-container deployment.

- Healthcheck = startup ordering.
- Internal network = better security.
- Reverse proxy = routing.
- `podman generate kube` exports Kubernetes YAML.
- `podman kube play` deploys Kubernetes YAML.